

$$\text{Sigmoid function} = \frac{1}{1+e^{-z}}$$

$$a = \frac{1}{1+e^{-z}}, \text{ prediction} = a$$

$$L = (a-y)^2 \text{ where } y = \text{target}$$

Differentiating the sigmoid

$$a = (1+e^{-z})^{-1}$$

$$\frac{da}{dz} = -(1+e^{-z})^{-2} \cdot (-e^{-z})$$

$$= \frac{e^{-z}}{(1+e^{-z})^2}$$

$$1-a = 1 - \frac{1}{1+e^{-z}} = \frac{e^{-z}}{1+e^{-z}}$$

Therefore,

$$\frac{da}{dz} = \frac{e^{-z}}{1+e^{-z}} \cdot \frac{1}{1+e^{-z}}$$

$$\frac{da}{dz} = a(1-a)$$

Now differentiate the loss $= \frac{1}{1+e^{-z}}$ = sigmoid function

$$L = (a-y)^2$$

$$\frac{dL}{da} = 2(a-y)$$

Applying the chain rule

$$\frac{dL}{dz} = \frac{dL}{da} \cdot \frac{da}{dz}$$

Substituting the result

$$\frac{dL}{dz} = 2(a-y) \cdot a(1-a)$$

$$\frac{dL}{dz} = 2(a-y)a(1-a)$$

$$= 2 \times \text{prediction} \times (1 - \text{prediction})$$